Examiner's Use

3 **(a)** As seen in Fig 3.1, a monochromatic light of wavelength 580 nm is used to produce an interference pattern on screen AB. The separation between the slits is 0.41 mm and the perpendicular distance between the double slit and the screen is *D*. Point Y is at a distance *x* = 2 mm from point O and it is the position of the first dark fringe. The intensity of the light passing through the two slits is the same.

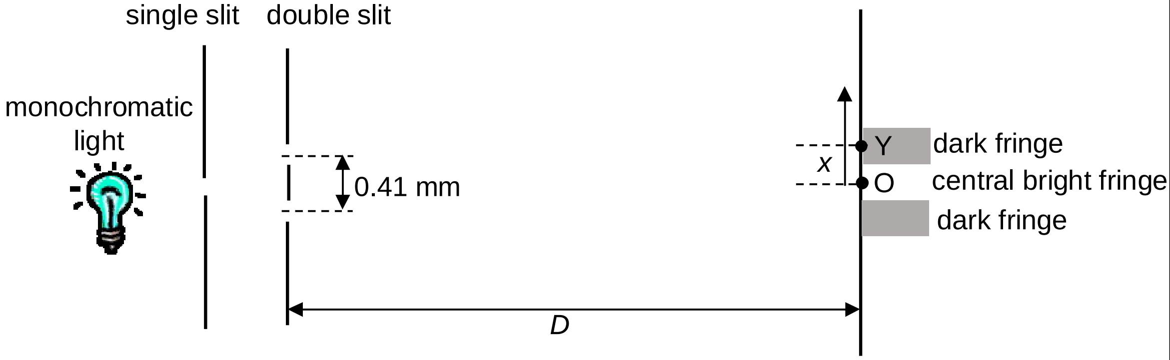

Fig 3.1

(i) Calculate the path difference between the 2 waves arriving at point Y from the slits.

Path difference = λ/2 = 580 /2 = 290 nm

distance = ……………….. nm [1]

(ii) Calculate the distance *D*.

By using x = λD/a	
$D = 0.41 \times 10^{-3} (2 \times 2 \times 10^{-3}) / 580 \times 10^{-9}$	[M1]
= 2.8 m	[A1]

D = ……………………… m [2]

(ii) The width of both slits is reduced by the same amount without altering their separation. The original variation with distance *x* from point O of the intensity is as shown in Fig. 3.2. Sketch the new variation of intensity on Fig. 3.2. [3]

Peaks at same location [1] (YDS)
More peaks & and wider envelope (minima shift outwards) [1]
Lower peaks [1]

Not marking for first maxima details.

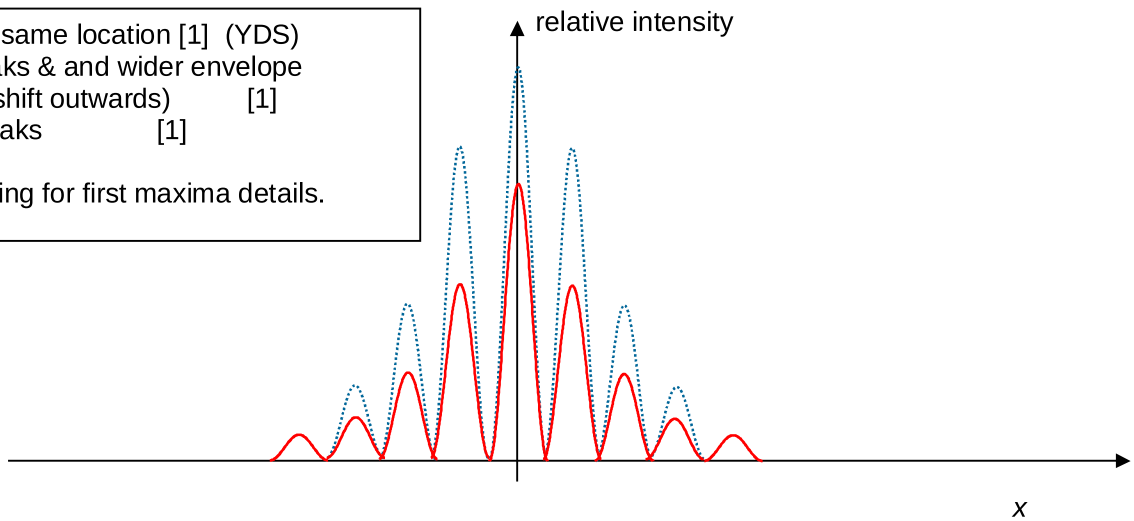

Fig. 3.2

For Examiner's Use

(b) A diffraction grating is used to measure the wavelengths of light. The angle θ of the second order maximum is measured for each wavelength. The variation with wavelength λ of sin θ is shown in Fig 3.3.

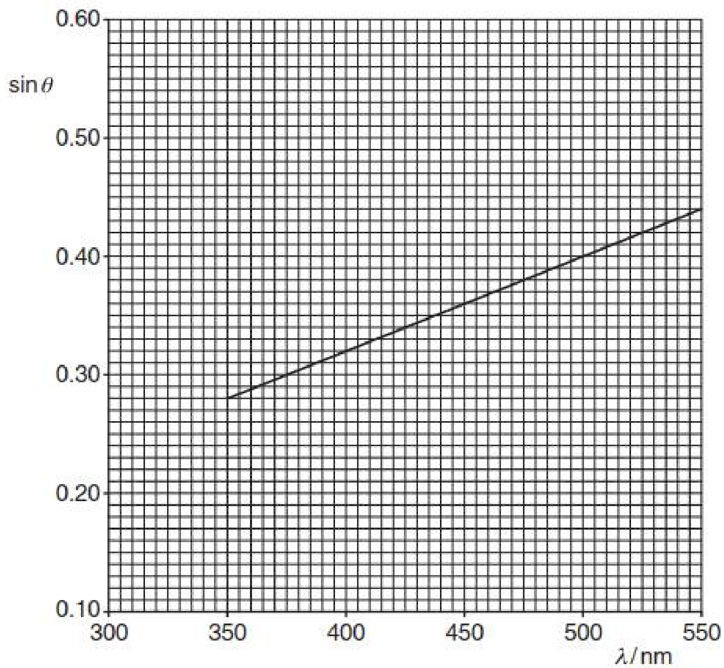

Fig. 3.3

(i) Calculate the slit separation *d* of the diffraction grating.

Gradient = 8.0×10^{5} [B1]

d = n / gradient = $2 / 8.0 \times 10^{5}$ [M1]

= 2.5×10^{-6} m [A1]

d = ……………………….m [3]

(ii) On Fig. 3.3, sketch a line to show the results that would be obtained for the first order maxima. [1]

straight line drawn with lower gradient (½) and all new y coordinates are ½ of the original y values

4 **(a)** Define *electric potential* at a point.